Lab Title: Design and configure a small LAN using Packet Tracer with at least 4 PCs, a switch, and a router. Assign IP addresses manually and test connectivity using ping.

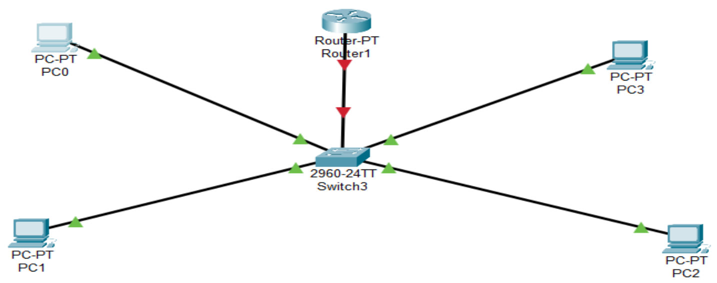
Objective:

To design and configure a small Local Area Network (LAN) consisting of 4 PCs, a switch, and a router. Assign IP addresses manually and test the connectivity using the ping command.

Apparatus/Software Required:

- Cisco Packet Tracer
- 4 PCs
- 1 Switch
- 1 Router
- Copper Straight-Through Cables

Network Topology Diagram:



IP Addressing Scheme:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet 0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	FastEthernet 0	192.168.10.11	255.255.255.0	192.168.10.1
PC2	FastEthernet 0	192.168.10.12	255.255.255.0	192.168.10.1
PC3	FastEthernet 0	192.168.10.13	255.255.255.0	192.168.10.1
Router	FastEthernet 0/0	192.168.10.1	255.255.255.0	-

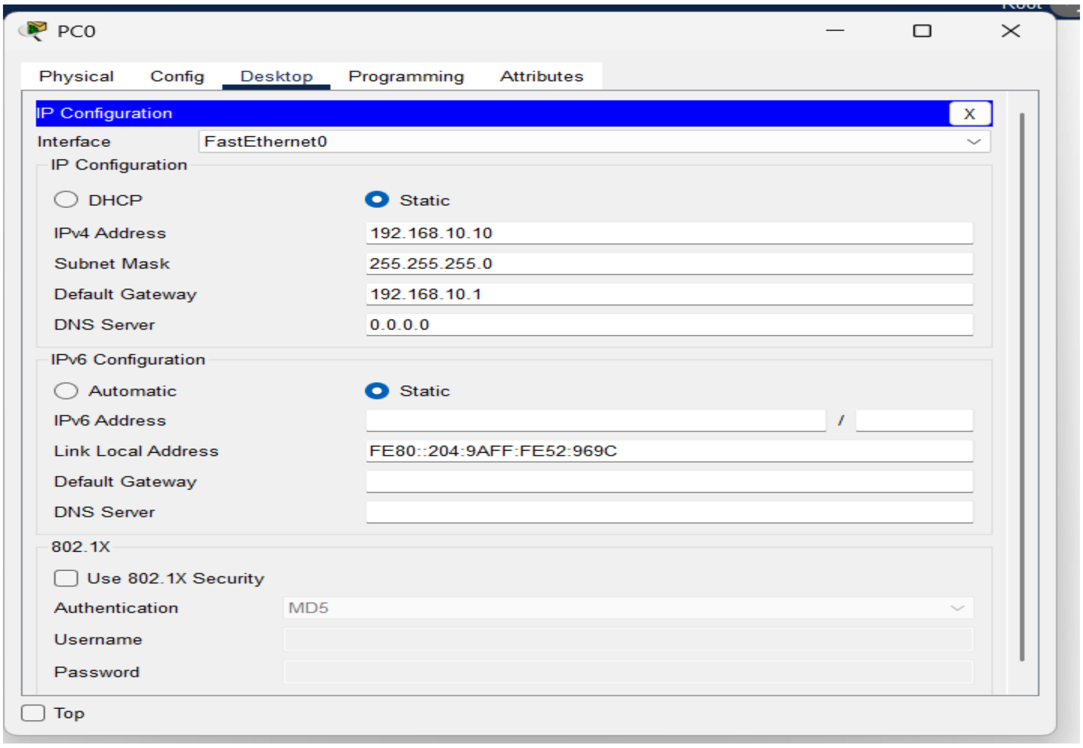
Procedure:

Step 1: Device Placement and Cabling

- Place 4 PCs, a Switch, and a Router in the workspace.
- Use Copper Straight-Through Cables to connect:
  - PCs to Switch
  - Switch to Router

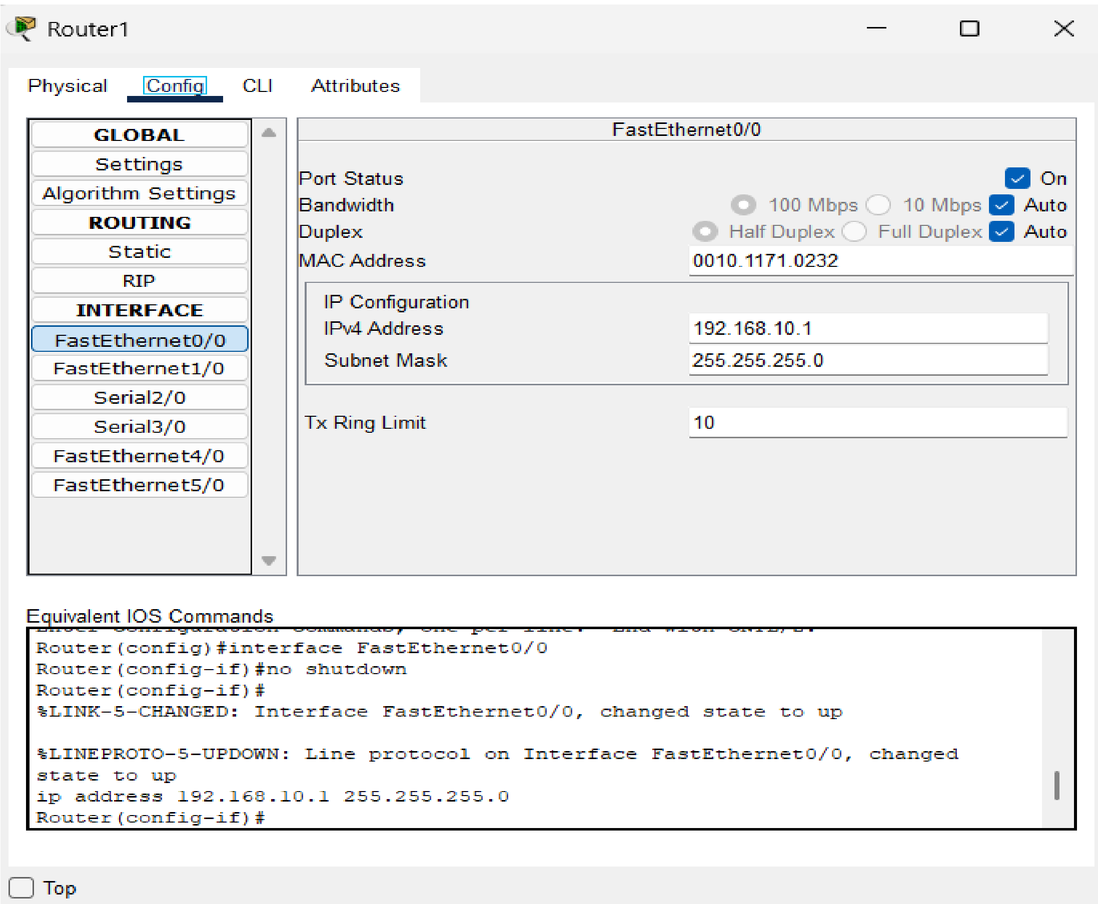
Step 2: IP Address Assignment on PCs

- Click on each PC → Desktop → IP Configuration.
- Enter the IP address, subnet mask, and default gateway as per the table above.



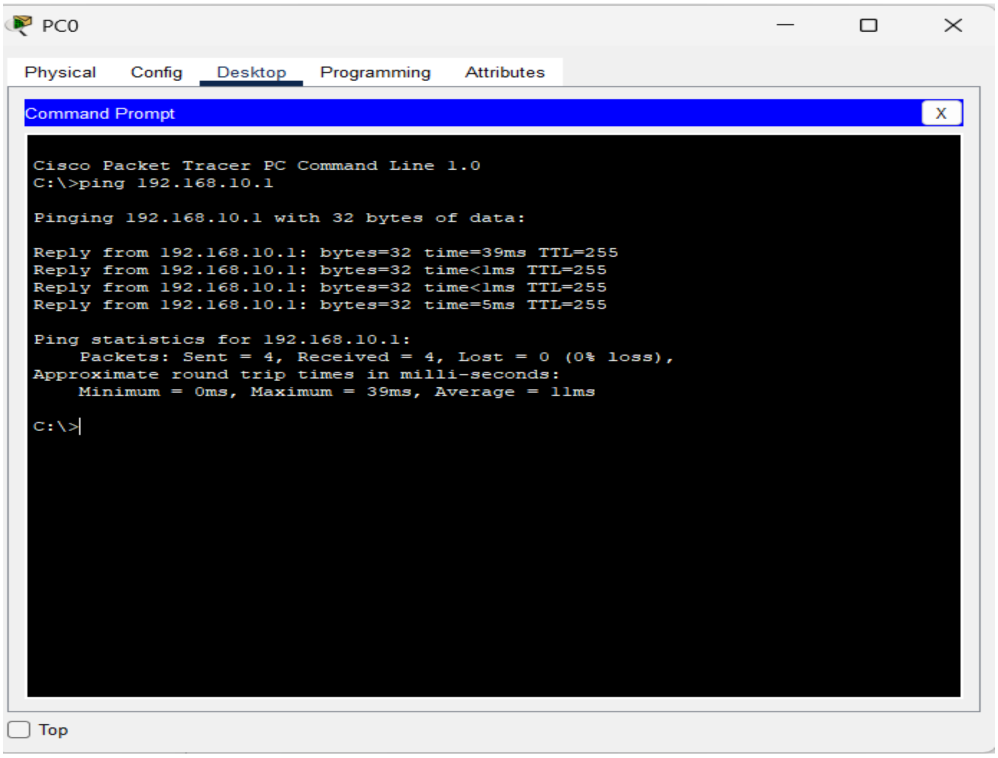
Step 3: Configure Router Interface

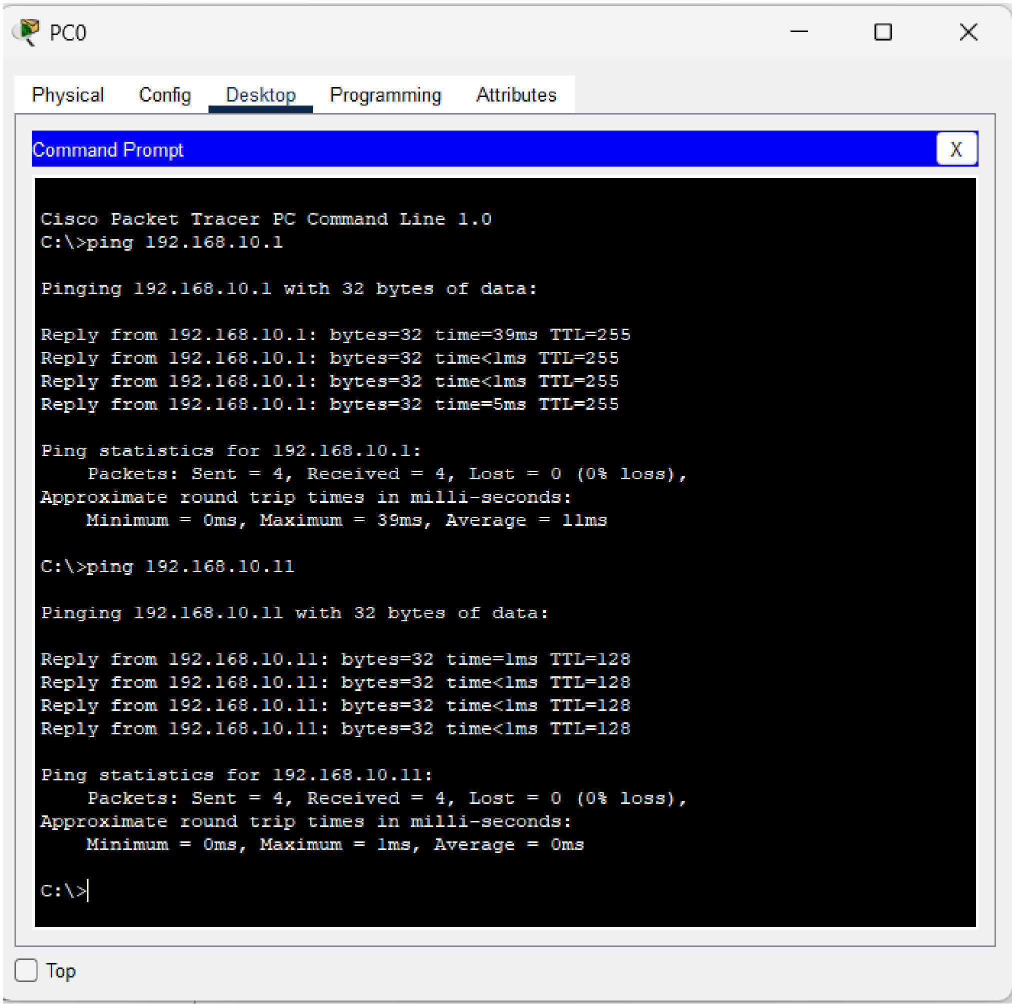
- Click the Router → Config → FastEthernet0/0



Step 4: Test Connectivity

- Open Command Prompt on each PC
- Use the ping command to test connectivity:





Observation:

Each PC successfully sends and receives ping responses to/from:

- The Router (192.168.10.1)
- Other PCs on the same network

Result:

A small LAN with 4 PCs, a switch, and a router was successfully designed and configured. Manual IP addressing was applied, and connectivity was verified using the ping command.

Conclusion:

This lab demonstrated the practical steps of creating a basic LAN, assigning static IP addresses, and testing communication across the network. This is foundational for understanding larger and more complex network topologies.